

Training and Testing Data

1. INTRODUCTION

In Machine Learning, we give data to a computer program so that it can **learn patterns** and make predictions.

But we cannot use all data only for learning. We need some data to **check whether the model has learned correctly or not**.

Therefore, we divide our dataset into two parts:

1. **Training Data**
 2. **Testing Data**
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2. TRAINING DATA

Definition

Training data is the data used to teach the machine learning model.

The model studies this data and learns relationships between input and output.

Example: Student Pass Prediction

Dataset:

Study Hours	Result
2	Fail
3	Fail
4	Pass
5	Pass
6	Pass
7	Pass

Suppose we give this data to the model:

Study Hours → Result

The model learns:

More study hours → Higher chance of passing

This learning process is called **Training**.

3. TESTING DATA

Definition

Testing data is the new data used to check how well the trained model performs.

The model has never seen this data before.

Example:

Training Data:

Study Hours Result

2 Fail

3 Fail

4 Pass

5 Pass

Testing Data:

Study Hours Result

6 Pass

1 Fail

The model predicts:

Study Hours = 6 → Pass

Study Hours = 1 → Fail

Then we compare:

Actual Result vs Predicted Result

4. WHY DO WE SPLIT DATA?

Suppose a student memorizes all answers from a book.

You ask the same questions.

Student gets:

100/100

Does it mean the student understands?

No.

The student only memorized.

Similarly, a machine learning model can memorize training data.

This problem is called:

Overfitting

5. OVERFITTING

Meaning

When a model performs very well on training data but performs poorly on new data.

Example:

Training Accuracy:

99%

Testing Accuracy:

60%

Meaning:

The model learned the training data too closely but cannot handle new data.

6. UNDERFITTING

Meaning

When a model does not learn enough from the data.

Example:

Training Accuracy:

55%

Testing Accuracy:

50%

The model is too simple and cannot find patterns.

7. TRAIN-TEST SPLIT

Usually, we divide data like:

Common Split

80-20 Rule

80% Data → Training

20% Data → Testing

Example:

1000 records:

800 records → Training

200 records → Testing

Other common splits:

Training Testing

70%	30%
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75%	25%
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80%	20%
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90%	10%
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8. IMPORTANT TERMS

Term	Meaning
Feature (X)	Input data used for prediction
Target (y)	Output we want to predict
Training Data	Data used for learning
Testing Data	Data used for checking performance
Model	Machine learning algorithm
Prediction	Output given by model
Accuracy	How many predictions are correct